

Rehabilitation and Burn Mass Casualty Incidents

Why rehabilitation must be part of every burn MCI response — from preparedness through long-term recovery.

Did you know?

Rehabilitation professionals play a critical role in burn recovery. Early rehabilitation is essential to prevent avoidable complications and improve long-term outcomes.

WHAT IS REHABILITATION IN BURN MASS CASUALTY INCIDENTS?

A mass casualty incident (MCI) occurs when the number or complexity of casualties exceeds the health resources available to provide necessary care. Burn MCIs occur when multiple people sustain burn injuries at the same time, often due to fires, explosions, or other hazards. Burn injuries are resource-intensive, and even relatively small numbers of casualties can place significant strain on health facilities. Coordinated multidisciplinary emergency responses are required to manage burn MCIs effectively.

Rehabilitation is an essential component of the response to burn MCIs. Early rehabilitation — beginning as soon as possible after injury — helps prevent avoidable complications, supports recovery, and improves long-term functioning outcomes.

WHAT IS THE CURRENT SITUATION?

Burn MCIs occur in many contexts including extreme weather events, armed conflict, industrial accidents, and large structural or wild fires.

- Large fires in densely populated settings, including refugee camps and informal settlements, can lead to burn MCIs.
- Fuel tanker explosions and transport accidents are a frequent cause of burn MCIs in low- and middle-income countries.
- Multiple burn MCIs have occurred as the result of fires in nightclubs all over the world, often caused by pyrotechnics.
- In 2019, the Whakaari / White Island volcanic eruption resulted in a large number of severely burned survivors, illustrating how burn MCIs can occur suddenly in diverse settings.
- Advances in burn care mean that more patients are surviving severe burn injuries, increasing the need to address long-term morbidity and functioning outcomes.
- Despite the critical role of rehabilitation in burn recovery, rehabilitation input is often delayed or overlooked during MCIs, particularly when health systems and referral pathways are disrupted.



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WHAT IS THE ROLE OF REHABILITATION IN BURN MASS CASUALTY INCIDENTS?

Rehabilitation plays a critical role in optimising outcomes following burn injuries and should begin as early as possible after injury. Timely rehabilitation for burn MCI survivors reduces demand for ongoing health services and supports survivors to return to education, livelihoods and meaningful activities.

Delayed or inadequate rehabilitation can lead to avoidable complications including:

- Joint contractures and restricted mobility
- Chronic pain and reduced quality of life
- Respiratory complications
- Increased need for corrective surgery
- Reduced functional independence and participation

Rehabilitation professionals provide essential interventions to support physical, psychosocial, and functional recovery. These may include:

- Scar management and anti-contracture positioning
- Splinting and orthotic provision
- Respiratory therapies and airway management support
- Mobility and strength exercises
- Swallowing and speech therapies
- Psychosocial support and adjustment counselling
- Functional retraining to restore independence in daily activities

WHAT ACTIONS ARE NEEDED?

- Include rehabilitation professionals in burn MCI response planning, training and preparedness activities.
- Ensure rehabilitation professionals responding to burn MCIs receive specialised training in burn care, including scar management, contracture prevention and the psychological impacts of burn injury.
- Plan for the long-term nature of burn rehabilitation: health services must ensure rehabilitation is adequately resourced throughout the full duration of survivors' recovery, which may extend months or years beyond the initial incident.
- Ensure rehabilitation services are activated at the commencement of the response to burn MCIs as a standard component of the health response.
- Integrate rehabilitation into burn-specific emergency supply chain planning to ensure essential equipment — including splinting materials, compression garments and mobility aids — is available from the earliest stages of response.
- Raise public awareness about burn prevention and correct burn first aid to reduce the incidence and severity of injuries.

About the World Rehabilitation Alliance

The World Rehabilitation Alliance is a WHO-hosted global network of more than 100 member organisations advancing rehabilitation in health systems and the recognition of functioning as a dimension of population health. Anchored in the Rehabilitation 2030 Initiative and WHA Resolution 76.6 (2023), the Alliance convenes partners across health, policy, and the wider development community.



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